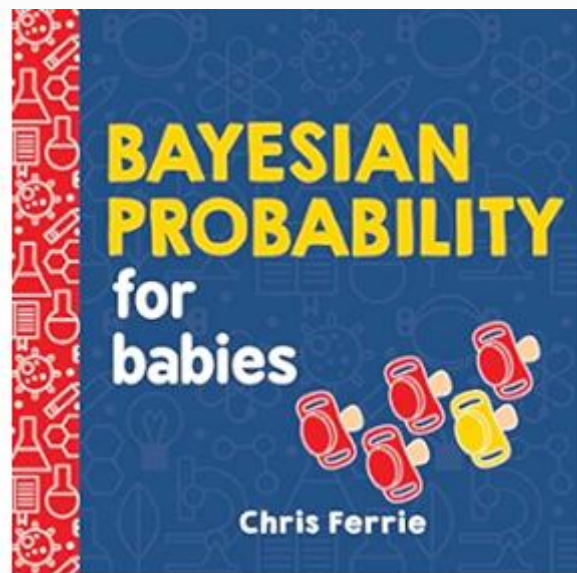




If the cookie had candy, then very few bites would have no candy.

$$\Pr(\text{no candy bite} \mid \text{candy cookie}) = \frac{1}{3}$$

The probability of a no-candy bite, given a candy cookie, is 1/3.



If the cookie had no candy, then every bite would have no candy.

$$\Pr(\text{no candy bite} \mid \text{no candy cookie}) = 1$$

The probability of a no-candy bite, given a no-candy cookie, is 1.

CS 1671 / CS 2071

Human Language Technologies

Session 4: Probability and linear algebra review

Michael Miller Yoder

September 3, 2026



University of
Pittsburgh

School of Computing and Information

Overview: Linear algebra and probability review

1. Preprocessing coding activity
2. Probability review
3. Linear algebra review

Copying the class Python environment

After installing Pixi, download the **new** pyproject.toml file

- Create a new directory called whatever you like
 - `mkdir class-env`
 - `cd class-env`
- Download the ***new*** manifest file that lists all packages in the class environment
 - I removed the problematic 'huggingface' channel
 - [pyproject.toml](#)
 - Alternative link:
https://github.com/michaelmilleryoder/cs1671_fall2026_jupyterhub/blob/main/pyproject.toml

Install packages

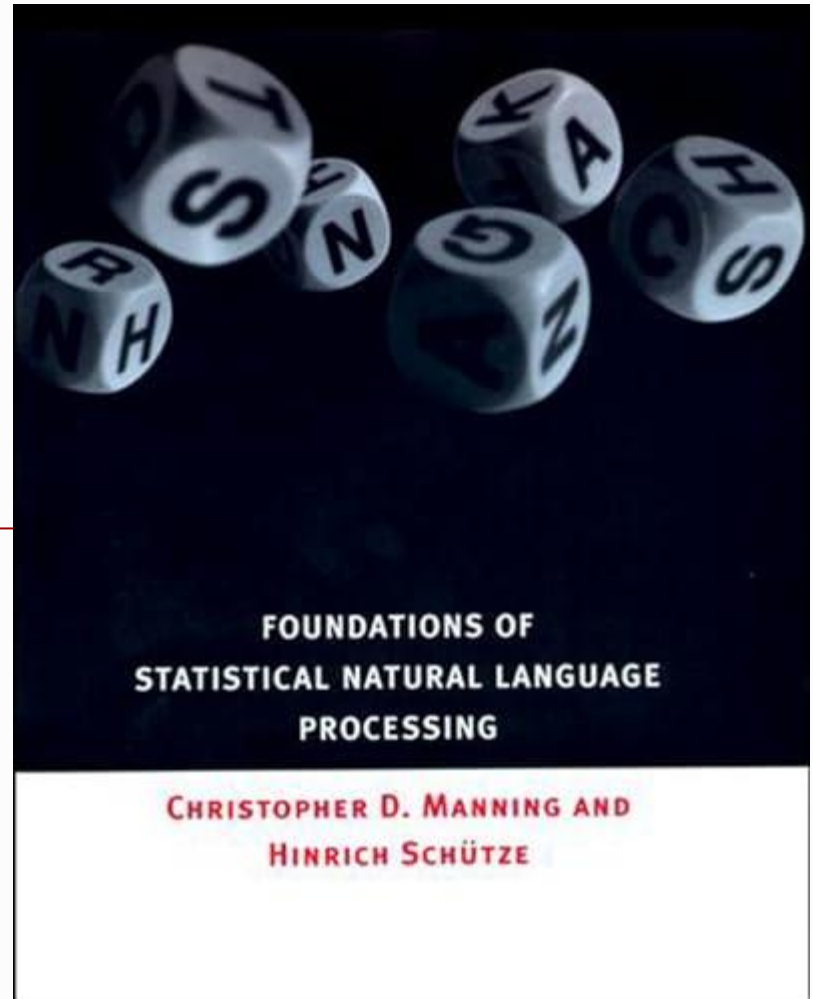
- Install packages in the class environment
 - `pixi install`
- This will take awhile! In the meantime, you can view what's in the `pyproject.toml` “manifest” file in a text editor
- Test whether you have the class environment installed
 - `pixi run python --version`
 - This should display Python 3.11.16

Coding activity: Preprocessing Airbnb listings

Coding activity: preprocessing Airbnb listings

1. Go to this [nbgitpuller link](#) (also available on course website)
2. Log in with your Pitt username if necessary
3. Start a server with **TEACH – 6 CPUs, 48 GB**
4. Load custom environment at
`/ix1/cs1671-2026f/class-env/.pixi/envs/default`
5. This should pull the `cs1671_fall2026_jupyterhub` folder into your JupyterLab
6. Open `session3_preprocessing.ipynb`

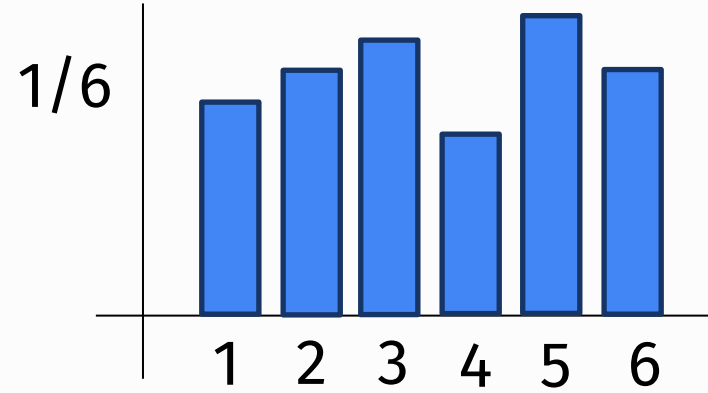
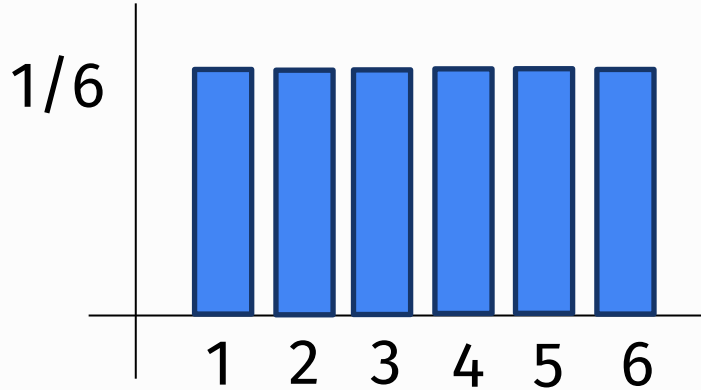
Probability review



Probability

- Probability of an event a occurring
- $P(a)$
 - For example, a could be a die showing a 2 out of $\{1, 2, 3, 4, 5, 6\}$
- Estimate $P(a)$ as $\frac{\text{count}(a)}{\text{count}(\text{all events})}$
 - Relative frequency or maximum likelihood estimate (MLE)

Probability distributions



Random variables

- **Random variable:** a variable that can hold multiple values according to a probability distribution
 - Probability distribution of values can be estimated empirically
 - For example, flipping a coin multiple times (possible outcomes {H, T}) and recording the result as 0 for tails and 1 for heads
- Distribution of a random variable X
 - $P(X)$ is a probability distribution over all possible values in the sample space. Probability mass function
 - $P(X = x)$ is the probability that the random variable X has the value x
 - $P(X = \text{heads})$, where X is the random variable of a coin flip

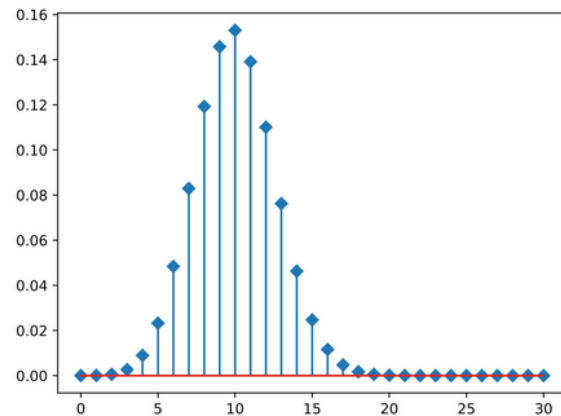


Figure 7.1: $P(k \text{ heads})$ in 30 tosses, success prob $1/3$.

Joint probability

- Probability of 2 events both occurring

$$P(A \cap B)$$

$$P(A, B)$$

- When rolling 2 dice, what's the probability of getting two 5s?

Let D_1 be dice 1, D_2 be dice 2. These events are independent, so:

$$P(D_1 = 5, D_2 = 5) = P(D_1 = 5) \cdot P(D_2 = 5)$$

$$\frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} \text{ since there are 36 different possible combinations}$$

Conditional probability

- Probability distributions sometimes change if you know another event has occurred or not occurred
- **Conditional** probability of an event a occurring given that another event, b , has already occurred
 - $P(a|b)$

Conditional probability

- Assume
 - X is the outcome of rolling a die once
 - F is the event $X = 6$
 - E is the event $X > 4$
- Die is rolled and we are told that E has occurred
- What is $P(F|E)$, that is, $P(X=6|X>4)$?

Conditional probability

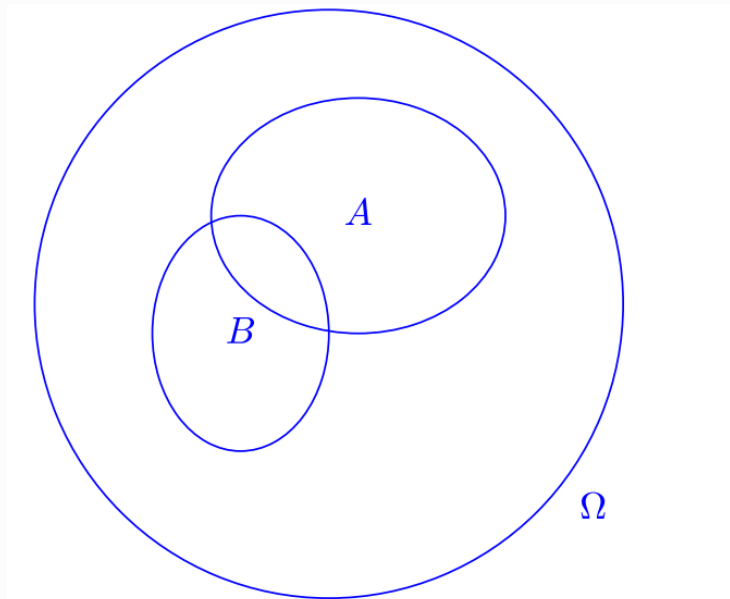


Figure 4.1: Events on the dart board

- Assume a very bad dart thrower (maybe Michael)

$$\mathbf{P}(A) = \frac{\mathbf{area}(A)}{\mathbf{area}(\Omega)}$$

Conditional probability

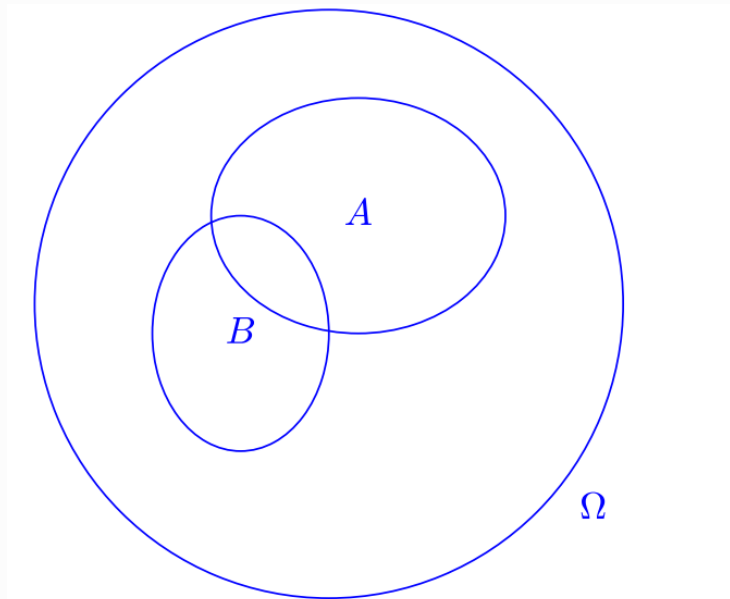


Figure 4.1: Events on the dart board

- You don't see the throw, but somebody tells you that the dart landed in B (so B occurred)
- What is the formula for $P(A|B)$?

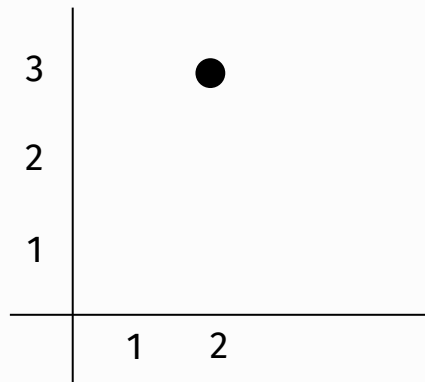
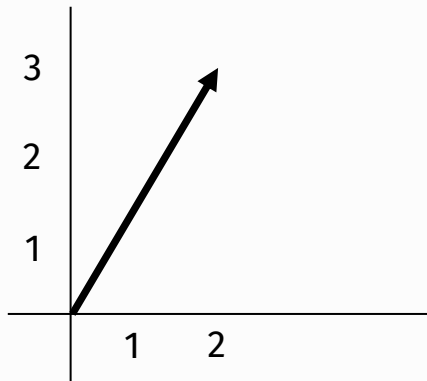
Linear algebra review

Vectors

An array of numbers with D dimensions

[2 3]

can be represented as a point in D -dimensional space



Dot product: vector $\cdot$ vector

Sum of the products of each vector dimension

$$\begin{array}{c} \mathbf{v} \end{array} \begin{array}{cccc} v_1 & v_2 & \cdots & v_N \end{array} \cdot \begin{array}{c} \mathbf{w} \\ w_1 \\ w_2 \\ \vdots \\ w_N \end{array}$$

$$\mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^N v_i w_i = v_1 w_1 + v_2 w_2 + \cdots + v_N w_N$$

Matrices

A matrix is an array of numbers

$$\begin{bmatrix} 6 & 4 & 24 \\ 1 & -9 & 8 \end{bmatrix}$$

Two rows, three columns.

It's Easy to Multiply a Matrix by a Scalar

$$2 \cdot \begin{bmatrix} 5 & 2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 \cdot 5 & 2 \cdot 2 \\ 2 \cdot 3 & 2 \cdot 1 \end{bmatrix} = \begin{bmatrix} 10 & 4 \\ 6 & 2 \end{bmatrix}$$

Dot product: vector $\cdot$ matrix

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \\ \\ \end{bmatrix}$$

Dot product: matrix · matrix

Let a_1 and a_2 be the row vectors of matrix A and b_1 and b_2 be the column vectors of a matrix B . Find $C = AB$

$$\begin{bmatrix} \boxed{1} & \boxed{7} \\ \boxed{2} & \boxed{4} \end{bmatrix} \cdot \begin{bmatrix} \boxed{3} & \boxed{3} \\ \boxed{5} & \boxed{2} \end{bmatrix} = \begin{bmatrix} \boxed{a_1 \cdot b_1} & \boxed{a_1 \cdot b_2} \\ \boxed{a_2 \cdot b_1} & \boxed{a_2 \cdot b_2} \end{bmatrix} = \begin{bmatrix} \boxed{} & \boxed{} \\ \boxed{} & \boxed{} \end{bmatrix}$$

A must have the same number of rows as B has columns.